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Asymptotic Analysis of Some Sequences

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In this section, we analyze the asymptotic behavior of lower bounds of $\mathbb{E}[Y_n]$, as follows.

$$x_{n+2} = \frac{a}{n} \sum_{i=1}^n x_i + b, \quad n \geq 1, \quad x_1 = x_2 = 0. \quad (1.1)$$

$$x_{n+2} = \frac{a}{n} \sum_{i=1}^n x_i + \frac{b}{n} \sum_{i=1}^n |x_i - x_{n+1-i}| + c, \quad n \geq 1, \quad x_1 = x_2 = 0, \quad (1.2)$$

where $a > 1, 0 < b < 1$ and $c > 0$.

$$x_n = \frac{2u}{n} \sum_{i=0}^{n-1} x_i + \frac{2v}{n} \sum_{i=\lfloor (n-1)/2 \rfloor}^{n-1} x_i, \quad n \geq 2, \quad x_0 = x_1 = 1, \quad (1.3)$$

where $u, v > 0$ and $u + v \leq 1 < 2u + v$.

Note. I shifted the index wrongly in recurrence (1.3) in the previous version. The correct version is as below. Due to experiments, the asymptotic behavior of the two are the same, but maybe the induction proofs differ.

$$x_{n+2} = \frac{2u}{n} \sum_{i=1}^n x_i + \frac{2v}{n} \sum_{i=\lfloor (n+1)/2 \rfloor}^n x_i, \quad n \geq 2, \quad x_1 = x_2 = 1, \quad (1.4)$$

where $u, v > 0$ and $u + v \leq 1 < 2u + v$.

Recurrence (1.4) is in turn a lower bound of the following, when we ignore the mid term in the sum of maxima.

$$x_{n+2} = \frac{2u}{n} \sum_{i=1}^n x_i + \frac{v}{n} \sum_{i=1}^n \max(x_i, x_{n+1-i}), \quad n \geq 2, \quad x_1 = x_2 = 1, \quad (1.5)$$

where $u, v > 0$ and $u + v \leq 1 < 2u + v$.

We can only lower-bound (1.5) by (1.4) if we can show that the sequence define by (1.5) is increasing.

Proposition 1. *Consider recurrence (1.1). We have*

- If $a < 1$, then $x_n \sim \frac{b}{1-a}$.
- If $a = 1$, then $x_n \sim b \ln n$.
- If $a > 1$, then $x_n \sim \frac{b}{(a-1)\Gamma(a)} n^{a-1}$.

Proof. Consider the generating function $X(z) = \sum_{n=1}^{\infty} x_n z^n$ with the convention that $x_0 = 0$. We rewrite the recurrence as

$$(n+2)x_{n+2} - 2x_{n+2} = a \sum_{i=1}^n x_i + bn.$$

We multiply both sides by z^{n+2} and sum over $n \geq 1$. The left-hand side becomes

$$\begin{aligned} \sum_{n=1}^{\infty} ((n+2)x_{n+2} - 2x_{n+2})z^{n+2} &= \sum_{n=3}^{\infty} nx_n z^n - 2 \sum_{n=3}^{\infty} x_n z^n \\ &= z \sum_{n=1}^{\infty} nx_n z^{n-1} - 2 \sum_{n=1}^{\infty} x_n z^n \\ &= zX'(z) - 2X(z). \end{aligned}$$

The right-hand side becomes

$$\begin{aligned} \sum_{n=1}^{\infty} (a \sum_{i=1}^n x_i + bn)z^{n+2} &= az^2 \sum_{n=1}^{\infty} \sum_{i=1}^n x_i z^n + bz^2 \sum_{n=1}^{\infty} nz^n \\ &= az^2 \frac{X(z)}{1-z} + \frac{bz^3}{(1-z)^2}. \end{aligned}$$

Therefore, the generating function $X(z)$ satisfies the following ODE

$$X'(z) - \left(\frac{2}{z} + \frac{az}{1-z} \right) X(z) = \frac{bz^3}{(1-z)^2}.$$

We have $\mu(z) = \exp \left(\int - \left(\frac{2}{z} + \frac{a}{1-z} - a \right) dz \right) = z^{-2}(1-z)^a$. which gives

$$z^{-2}(1-z)^a X(z) = \int b(1-z)^{a-2} dz.$$

There are two cases of a that determine the integral. If $a = 1$, we have

$$z^{-2}(1-z)X(z) = b \ln \left(\frac{1}{1-z} \right) + C.$$

Since $x_1 = x_2 = 0$, we have $z^{-2}X(z) \rightarrow 0$ as $z \rightarrow 0$, which gives $C = 0$. Therefore, we have

$$X(z) = b \frac{z^2}{1-z} \ln \left(\frac{1}{1-z} \right).$$

Hence, $x_n = bH_{n-2} \sim b \ln n$.

If $a \neq 1$, we have

$$z^{-2}(1-z)^a X(z) = -\frac{b}{a-1} (1-z)^{a-1} + C.$$

Using the same argument as above, we have $C = \frac{b}{a-1}$. Therefore, we have

$$X(z) = \frac{b}{a-1} z^2 \left((1-z)^{-a} - (1-z)^{-1} \right).$$

Using the generalized binomial theorem

$$(1 - z)^{-a} = \sum_{n=0}^{\infty} \binom{n + a - 1}{n} z^n,$$

we have

$$x_n = \frac{b}{a-1} \left(\binom{n-3+a}{n-2} - 1 \right) \sim \frac{b}{a-1} \left(\frac{n^{a-1}}{\Gamma(a)} - 1 \right).$$

If $a < 1$ then $x_n \sim \frac{b}{1-a}$. If $a > 1$, then $x_n \sim \frac{b}{(a-1)\Gamma(a)} n^{a-1}$. □

Lemma 1. *The sequence (x_n) in (1.2) is increasing.*

Proof. One checks that $x_n \geq 0$ for all n . Let us prove that (x_n) is increasing. We have $x_2 \geq x_1$. Assume that $x_n \geq x_{n-1} \geq \dots \geq x_0$ for some $n \geq 3$. Then,

$$\begin{aligned} nx_{n+2} &= a \sum_{i=1}^n x_i + b \sum_{i=1}^n |x_i - x_{n+1-i}| + nc \\ &= a \sum_{i=1}^n x_i + b \sum_{i=1}^{\lfloor n/2 \rfloor} (x_{n+1-i} - x_i) + nc \\ &= a \sum_{i=1}^n x_i + b \sum_{i=1}^{\lfloor n/2 \rfloor} (x_{n+1-i} - x_i) + nc \\ &= a \sum_{i=1}^n x_i + b \sum_{i=1}^{\lfloor n/2 \rfloor} (x_{n+1-i} - x_i) + nc \\ &= a \sum_{i=1}^n x_i + b \sum_{i=1}^{\lfloor n/2 \rfloor} (x_{n+1-i} - x_i) + nc \\ &= (a+b) \sum_{i=1}^n x_i - 2b \sum_{i=1}^{\lfloor n/2 \rfloor} x_i - bm_n + nc, \end{aligned}$$

where

$$m_n = \begin{cases} x_{\lfloor n/2 \rfloor + 1} & \text{if } n \text{ is odd,} \\ 0 & \text{if } n \text{ is even.} \end{cases}$$

Let $S_n = \sum_{i=1}^n x_i$. Then,

$$\begin{aligned} nx_{n+2} &= (a+b)S_n - 2bS_{\lfloor n/2 \rfloor} - bm_n + nc, \\ S_{n+1} &= S_n + x_{n+1}. \end{aligned}$$

Hence,

$$nx_{n+2} - (n-1)x_{n+1} = (a+b)x_n - 2b(S_{\lfloor n/2 \rfloor} - S_{\lfloor (n-1)/2 \rfloor}) - b(m_n - m_{n-1}) + c.$$

Let us check the right-hand side according to the parity of n .

- If $n = 2k$, then $S_{\lfloor n/2 \rfloor} - S_{\lfloor (n-1)/2 \rfloor} = x_k$ and $m_n - m_{n-1} = -x_k$. The right-hand side is $(a+b)x_n - 2bx_k + c$.

- If $n = 2k + 1$, then $S_{\lfloor n/2 \rfloor} - S_{\lfloor (n-1)/2 \rfloor} = 0$ and $m_n - m_{n-1} = x_{k+1}$. The right-hand side is $(a + b)x_n - 2bx_{k+1} + c$.

Collecting the two cases, we have

$$nx_{n+2} - (n - 1)x_{n+1} = (a + b)x_n - 2bx_{\lfloor n/2 \rfloor} + c.$$

The proof follows from Proposition ??.

□

Lemma 2. *The sequence in (1.2) can be rewritten as*

$$nx_{n+2} - (n - 1)x_{n+1} = (a + b)x_n - 2bx_{\lfloor n/2 \rfloor} + c. \quad (1.6)$$

Proposition 2. *Consider the sequence defined by (1.3). We have $x_n = \Theta(n^\alpha)$, where α is the unique solution of*

$$2(u + v) - v2^{-\alpha} = \alpha + 1.$$

Proof. For the lower bound, assume the inductive hypothesis $x_n \geq K(n^\alpha + 1)$ with $K \leq \frac{1}{2}$ to be determined. The base cases $n = 0, 1$ hold. Let us check for $n = 2$.

$$x_2 = 2u + 2v = \alpha + 1 + v2^{-\alpha} \geq \alpha + 1 \geq \frac{1}{2}(2^\alpha + 1), \forall \alpha \in [0, 1].$$

For $n \geq 3$, we have

$$\begin{aligned} \frac{x_n}{K} &\geq \frac{2u}{n} \sum_{i=0}^{n-1} (i^\alpha + 1) + \frac{2v}{n} \sum_{i=\lfloor (n-1)/2 \rfloor}^{n-1} (i^\alpha + 1) \\ &\geq \frac{2u}{n} \left(\left(\int_0^{n-1} x^\alpha dx \right) + n \right) + \frac{2v}{n} \int_{(n-3)/2}^{n-1} (x^\alpha + 1) dx \\ &= \frac{2u}{n} \left(\left(\int_0^{n-1} x^\alpha dx \right) + n \right) + \frac{2v}{n} \left(\int_{(n-3)/2}^{n-1} x^\alpha dx + \frac{n+1}{2} \right) \\ &= \frac{2u}{n} \left(\frac{(n-1)^{\alpha+1}}{\alpha+1} \right) + \frac{2v}{n} \left(\frac{(n-1)^{\alpha+1} - ((n-3)/2)^{\alpha+1}}{\alpha+1} \right) + 2u + v \left(1 + \frac{1}{n} \right) \\ &\geq \frac{(n-1)^{\alpha+1}}{n(\alpha+1)} \left(2u + 2v - v \left(\frac{n-3}{n} \right)^{\alpha+1} 2^{-\alpha} \right) + 2u + v \\ &\geq n^\alpha \left(1 - \frac{1}{n} \right)^{\alpha+1} + 2u + v \end{aligned}$$

We are done if we can prove that for every $n \geq 3$,

$$n^\alpha \left(1 - \frac{1}{n} \right)^{\alpha+1} + 2u + v \geq n^\alpha + 1.$$

By Bernoulli's inequality, we have $\left(1 - \frac{1}{n} \right)^{\alpha+1} \geq 1 - \frac{\alpha+1}{n}$. Hence, it suffices to show that

$$n^\alpha \left(1 - \frac{\alpha+1}{n} \right) + 2u + v \geq n^\alpha + 1 \iff 2u + v - 1 \geq \frac{\alpha+1}{n^{1-\alpha}}.$$

Let $N_0 = \left\lceil \left(\frac{\alpha + 1}{2u + v - 1} \right)^{\frac{1}{1-\alpha}} \right\rceil$. The inequality holds for every $n \geq N_0$. Now we choose K for it to hold for $n < N_0$. In particular,

$$K = \min \left(\frac{1}{2}, \min_{3 \leq i < N_0} \frac{x_i}{i^\alpha + 1} \right) > 0.$$

The lower bound follows. □

Proposition 3. *Consider the sequence defined by (1.4). We have $x_n = \Theta(n^\alpha)$, where α is the unique solution of*

$$2(u + v) - v2^{-\alpha} = \alpha + 1.$$

Proof. The hypothesis for the lower bound is $x_n \geq K(n^\alpha + L)$, where K, L will be chosen to satisfy the base cases, which are

$$K(L + 1) \leq 1 \quad \text{and} \quad \alpha + 1 \geq K(3^\alpha + L).$$

Similar to the proof of the previous proposition, we have to show that for every $n \geq 3$,

$$n^\alpha + L(2u + v) \geq (n + 2)^\alpha + L \iff (n + 2)^\alpha - n^\alpha \leq L(2u + v - 1).$$

With $x > 0$ and $\alpha \in [0, 1]$, the function $x \mapsto (x + 2)^\alpha - x^\alpha$ is non-increasing. Hence, it suffices to show that

$$3^\alpha - 1 \leq L(2u + v - 1).$$

Now we choose

$$L = \frac{2}{2u + v - 1} \quad \text{and} \quad K = \frac{2u + v - 1}{2u + v + 1}.$$

Note that the optimal choice is $L = \frac{1}{K} - 1$. We have $K \leq \frac{1}{2}$. We have to show that the smaller K is, the easier it is to satisfy the base cases. One checks that

$$\alpha + 1 \geq \frac{1}{2}(3^\alpha + 1)$$

and

$$f(x) = \frac{1}{x}(3^\alpha + x - 1) = 1 + \frac{3^\alpha - 1}{x}$$

is decreasing when $x \geq 2$. □

The Yield Recurrence

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The recurrence we consider in this section is the following.

$$Y_{n+2} \stackrel{d}{=} I^{(n)} \left(Y_{U_n}^{(1)} + Y_{n+1-U_n}^{(2)} + 1 \right) + J^{(n)} \max \left(Y_{U_n}^{(1)}, Y_{n+1-U_n}^{(2)} \right). \quad (2.1)$$

Let us explain the notations. Let Y_n be the yield of the tree with n nodes. The variables I and J are Bernoulli with expectation u and v respectively, such that $I + J \leq 1$. That means we can only either take the sum-plus-one term, or the max term, or zero. The variable U_n is uniformly distributed on $\{1, \dots, n\}$. Superscripts are for independent copies. We assume that all collections of copies are independent. Now we can understand why it is correct to reuse the same variables $Y_{U_n}^{(1)}, Y_{n+1-U_n}^{(2)}$ and U_n across both terms.

2.1 Lower and Upper Bounds of the Expectation

Let $y_n = \mathbb{E}[Y_n]$. Take expectation on both sides, we have

$$y_{n+2} = u + \frac{2u}{n} \sum_{i=1}^n y_i + \frac{v}{n} \sum_{i=1}^n \mathbb{E}[\max(Y_i, Y_{n+1-i})].$$

For an upper bound, using $\max(Y_i, Y_{n+1-i}) \leq Y_i + Y_{n+1-i}$, we get an upper bound sequence

$$y_{n+2} = u + \frac{2(u+v)}{n} \sum_{i=1}^n y_i, \quad y_1 = y_2 = 0. \quad (2.2)$$

There are several ways to get a lower bound. The first one is to use convexity of the absolute value function.

$$\begin{aligned} \mathbb{E}[\max(Y_i, Y_{n+1-i})] &= \mathbb{E} \left[\frac{1}{2} (Y_i + Y_{n+1-i} + |Y_i - Y_{n+1-i}|) \right] \\ &\geq \frac{1}{2} (\mathbb{E}[Y_i] + \mathbb{E}[Y_{n+1-i}]) + \frac{1}{2} |\mathbb{E}[Y_i] - \mathbb{E}[Y_{n+1-i}]| \\ &= \frac{1}{2} (y_i + y_{n+1-i} + |y_i - y_{n+1-i}|). \end{aligned}$$

This gives

$$y_{n+2} = u + \frac{2u+v}{n} \sum_{i=1}^n y_i + \frac{v}{2n} \sum_{i=1}^n |y_i - y_{n+1-i}|, \quad y_1 = y_2 = 0. \quad (2.3)$$

Another way is to use convexity of the max function.

$$\mathbb{E}[\max(Y_i, Y_{n+1-i})] \geq \max(\mathbb{E}[Y_i], \mathbb{E}[Y_{n+1-i}]) = \max(y_i, y_{n+1-i}) = y_{\max(i, n+1-i)}.$$

This gives

$$y_{n+2} = u + \frac{2u}{n} \sum_{i=1}^n y_i + \frac{v}{n} \sum_{i=1}^n y_{\max(i, n+1-i)}, \quad y_1 = y_2 = 0. \quad (2.4)$$

If we can prove that y_n is non-decreasing in n , then we can ignore the middle term in the sum of maxima, and get

$$y_{n+2} = u + \frac{2u}{n} \sum_{i=1}^n y_i + \frac{2v}{n} \sum_{i=\lfloor (n+1)/2 \rfloor}^n y_i, \quad y_1 = y_2 = 0. \quad (2.5)$$

The weakest lower bound is to replace the max by the average, which gives

$$y_{n+2} = u + \frac{2u+v}{n} \sum_{i=1}^n y_i, \quad y_1 = y_2 = 0. \quad (2.6)$$

When $2u+v > 1$, the true expectation is bounded between by polynomial of degrees $2u+v-1$ and $2(u+v)-1$. This case agrees with real data. We conjecture that the true expectation also grows polynomially. Details of the analysis of the deterministic recurrences are in Chapter 1.

2.2 Asymptotic of the Expectation